Semester 1 Capstone

Master Game Design Document

Instructor Reference & Target Spec - not distributed to students

VIDEO GAME CREATION

Working Title (finalize with students - naming it is itself a fun early production step):

How to Use This Document

This is the instructor-facing master design for the game every student builds in Semester 1 - it's the answer key, not a student handout. It complements Game_Design_Document_Template.docx (the student's own living notebook) rather than replacing it.

- This mirrors real pre-production: a high concept gets pitched, then locked into a full design, then sliced into a production roadmap - here, the roadmap is lesson-by-lesson instead of sprint-by-sprint.
- Each lesson's own GDD "Goals" table (starting where a feature is introduced) is a slice of the Feature-to-Lesson Map below. If the design changes here, re-slice any lessons it affects.
- Students only see the High Concept up front (Lesson 1's GDD) - the rest of this spec is intentionally revealed to them incrementally, feature by feature, the same way a real dev team learns a design through production rather than all at once.

High Concept

- **Genre:** Top-down arcade shooter, in the tradition of Space Invaders and Galaga.
- **One-Sentence Pitch:** Waves of enemies descend from above - move, dodge, and shoot your way through before they reach you.
- **Inspiration:** Classic arcade shooters (Space Invaders, Galaga) - chosen for minimal art/animation requirements and a core loop that scales cleanly from a one-lesson MVP to a full semester of features.
- **Tone:** Light, arcade-y, all-ages - no violence beyond abstract "invader" sprites, no lose-state punishment beyond "try again".

Core Gameplay

- **Core Mechanic:** Move along the bottom of the screen and shoot upward to destroy enemies before they reach you.

Controls

Input	Action
Left / Right Arrow	Move the player left or right along the bottom of the play area
Spacebar (or Up Arrow, if spacebar conflicts with an engine's own shortcuts)	Shoot a projectile upward

- **Win Condition:** Clear all enemies in the current wave (Level 1: a single wave. Level 2+: multiple waves.)
- **Lose Condition:** An enemy reaches the player's row, or the player runs out of lives (Level 2+).

World & Characters

- **Player:** A single ship or character confined to the bottom of the screen. No name/backstory needed - keep it abstract so every engine's default or placeholder assets work fine.
- **Enemies:** Simple, uniform "invader" enemies at Level 1. Level 3 may introduce a second enemy type or bonus enemy as an optional stretch.

Level 1 / 2 / 3 Feature Breakdown

Same three levels as every other lesson in this course - Level 1 is a real, complete finish, not a rough draft. Levels 2 and 3 build on it without changing what's already there.

Level 1 - MVP

- Move left/right via directional input
- Shoot a single projectile upward on a key press
- One fixed row of enemies present at start
- Projectile destroys an enemy on collision
- Win condition: all enemies destroyed

Level 2 - Complete

- Multiple rows / a full wave of enemies
- Enemies advance toward the player (lane-based descent - see Design Notes below)
- Lose condition: an enemy reaches the player, or the player is hit
- Player has limited lives (suggested: 3)
- Score tracking plus a basic UI (score, lives)
- Game-over / restart flow

Level 3 - Refined

Core Level 3 features (expected of everyone reaching this level):

- Multiple waves with increasing difficulty (more enemies, faster movement, or both)
- Sound effects: shoot, explosion, game over, wave clear
- Basic animation: explosion effect, hit flash/reaction

Optional stretch features (student's choice - this is where personalization lives):

- Destructible shields/barriers
- A power-up (rapid fire, spread shot, temporary shield)
- A bonus enemy worth extra points
- Classic formation-marching movement (side-to-side, dropping a row at each edge) instead of lane-based descent, for students ready to work with lists/arrays of enemies
- Title screen and/or game-over screen

Feature-to-Lesson Map

This is the slice-and-dice backbone: which capstone feature gets introduced in which lesson, matching the pacing already set by the 30-lesson milestone table in the Course Syllabus and GDD template. First draft - review and adjust before it's locked in, since every lesson built from here on inherits this pacing.

Lesson	Feature Introduced	Capstone Level
2	Player movement, left/right (this is Lesson 2's own Key Press Reactor goal - no separate capstone work needed, it's the same skill)	Level 1
3	Shooting: spawn a projectile, move it upward, remove it off-screen	Level 1
4	A single enemy on screen + collision detection (projectile destroys enemy)	Level 1
5	Multiple enemies (a row/list of enemies), loops to spawn and manage them	Level 1
6	Win condition: all enemies destroyed. Level 1 MVP milestone complete.	Level 1
7	Enemies advance toward the player (lane-based descent - see Design Notes)	Level 2
8	Lose condition: enemy reaches the player / player is hit, plus a lives system	Level 2
9	Score tracking and a basic on-screen UI (score, lives)	Level 2
10	Game-over and restart flow	Level 2
11	Multiple waves. Level 2 Complete milestone reached.	Level 2
12	Sound effects: shoot, explosion, game over, wave clear	Level 3
13	Basic animation: explosion effect, hit flash/reaction	Level 3
14	Difficulty scaling across waves; introduce optional stretch features	Level 3
15	Polish pass and Semester 1 Capstone presentation. Level 3 Refined milestone reached.	Level 3

Design Notes / Open Decisions

- **Direction-color feedback: sprite swap vs. color tint:** Decided (Lesson 2): color change will be implemented by sprite swapping - a separate pre-made colored sprite/costume per direction - rather than tinting one sprite at runtime with each engine's quick-color property (image_blend, modulate, spriteRenderer.color, Tint color, and similar). Tint was the original plan, but it only produces a clean, true color on white/grayscale source art; on our actual colored placeholder art it multiplies into a muddy result instead, discovered while live-testing the GameMaker path. Sprite-swap was chosen over the alternative fix (adding white art variants and keeping tint) for a few reasons: it's the simpler concept for this class's age/skill range - "show a different picture" needs no explanation of multiply-blend color math - it needed zero new assets (the placeholder pack already has one color per direction plus a spare for idle), and it works the same way in every engine with no exceptions. This is also a deliberate, admittedly somewhat arbitrary "faster vs. more sophisticated" call, the kind real studios make under time and team-skill constraints: tinting is more flexible long-term (any number of colors with no new art, smooth transitions possible) and stays available to any student ready for it as a Super-Stretch idea, once they understand why it needs white source art to look clean. TIC-80 is the deliberate exception, left on its original color-tint approach (poke4() palette remapping) rather than switched to sprite-swap - unlike every

other engine here, TIC-80's palette-based recolor is exact regardless of the source art's color, so it never had the muddy-blend problem sprite-swap exists to avoid; moving it to sprite-swap would trade away a technique that's already simpler and more idiomatic there for one it doesn't need.

- **Enemy movement: lane-based descent vs. formation marching:** Recommending lane-based descent (each enemy independently moves down its own lane) as the Level 1/2 default - it needs no shared group state and stays uniformly simple to build across all 8 engines, including block/visual tools and younger students. True Space-Invaders-style formation marching (shared movement, edge-bounce, row-drop, speed-up as enemies thin out) is offered instead as a Level 3 stretch feature, once students are comfortable with lists/arrays. Flagging for confirmation before this is locked into the Feature-to-Lesson Map above.
- **Working title:** Left blank on the cover - naming the game could even be a short class activity or vote once the High Concept is pitched in Lesson 1, mirroring how studios sometimes finalize a title well after a project is greenlit.